

District Demographics, Legislator Effectiveness, and Reelection in US State Legislatures

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Abstract

How do district income level and legislator occupational background and effectiveness interact to affect legislator reelection? I create an original dataset combining U.S. Census state legislative district per capita mean income with legislators' state legislative effectiveness score (SLES) and occupational background. The resulting dataset contains more than 10,000 observations for upper and lower chambers across 36 states for the years 2010-2016. I find that higher-income districts have more effective legislators, fewer working-class legislators, and are more likely to remove legislators, indicating that districts of different income levels behave differently with regard to their legislators. These findings have implications for redistricting and suggest that increased demographic data should be considered in the U.S. Census' standard redistricting file.

Keywords: state legislatures; representation; elections; state politics; legislative institutions; legislative effectiveness; income; economic class

Introduction

North Dakota's House of Representatives District 31 is a low-income district that covers about 4,267 square miles in the southeastern part of the state and includes the Cannonball Stage Station State Historic Site. It is represented by two legislators, one of whom has been Representative Karen M. Rohr since 2010. Representative Rohr is a board-certified Nurse Practitioner and adjunct Faculty Professor in Nursing with a PhD in Nursing. In the years from 2010 to 2016, she was the primary sponsor for only one bill, which didn't pass. Her State Legislative Effectiveness Score (SLES), calculated by the Center for Effective Lawmaking, has been low, labeling her an ineffective legislator, yet she has been reelected. Why have her constituents reelected her despite her ineffectiveness? How do her district's low-income level and her non-working-class background influence this female legislator's continued reelection?

Legislator effectiveness

Legislator effectiveness is a quantitative measure of how successful a legislator is at passing his or her sponsored bills or receiving appropriations. Jeydel and Taylor (2003) created their own measure based on legislative success and federal domestic funding appropriations using a federal spending database called the Federal Assistance Awards Data System from 1983 to 1997. About twenty

years later, Bucchianeri, Volden, and Wiseman (2024) created a newer measure based on legislative success at each step of the bill-enacting process; the more sponsored bills advance the farthest, the higher the legislative effectiveness score. The measure includes milestones such as “sponsorship, action in committee, action beyond committee, bill passing one chamber, and a bill becoming law” (Lollis 2024). This newer measure uses more detailed data and gives legislators credit for advancing bills into committee or to the floor even if they are not ultimately passed but doesn't take district appropriations into account - notable differences from Jeydel and Taylor's approach.

Descriptive representation

The creation of the SLES indicator has spurred research into descriptive representation. Of particular interest, Lollis (2024) finds that working-class legislators are not less effective than white-collar or elite legislators and suggests a positive selection bias as well as voter bias against working-class candidates in elections which result in more competent working-class legislators being elected. Working-class legislators behave differently than non-working-class legislators by being more focused on economic concerns (Carnes 2013), so their representation within legislatures has a substantive impact on legislative outcomes. I expand on the dataset and findings to investigate how district income level interacts with legislator occupational background and effectiveness to influence legislators' reelection.

Data and methods

I pair U.S. Census American Community Survey (ACS) Table S1902 per capita mean income data with SLES data from the Center for Effective Lawmaking (Bucchianeri, Volden, and Wiseman 2024) for the years 2010, 2012, 2014, and 2016 for state legislative districts for both upper and lower legislative chambers in 36 states (Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Michigan, Mississippi, Missouri, Montana, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Tennessee, Texas, Utah, Virginia, Washington, Wisconsin, and Wyoming). Various states were not included in the dataset due to missing data or inconsistent coding of district numbers. I use per capita mean income as an indicator of the district's socioeconomic level. Unfortunately, the U.S. Census ACS Table S1902 was not produced for years prior to 2010, which makes it difficult to investigate differences before and after a redistricting cycle. District numbers were filled in manually for 556 entries that were missing in the original SLES dataset. I also create a unique indicator variable for each state-chamber-term combination to

allow for fixed effects since the SLES measure is normalized within state-chamber-term. I then pair this dataset with Lollis' worker dataset (Lollis 2024), which provides working-class occupational classifications for state legislators. This resulting dataset contains 14,238 data points. I then create a new binary variable that indicates whether a legislator was reelected in the following election; the variable is 0 if the legislator was removed in the next election year and 1 if the legislator was reelected in the next election year. The completed dataset contains 10,321 data points.

Analysis



Figure 1: Stacked bar chart of District Per Capita Mean Income Level and SLES by Worker status and reelected status

Fixed effects

The SLES indicator is normalized by state, chamber, and term, so scores cannot be compared directly across states, chambers, or terms. I create a unique indicator variable concatenating state, chamber, and term to allow for fixed effects. This makes it impossible to use chamber or district population as an explanatory variable. Another approach could be to create binning variables for SLES and income, labeling districts' per capita mean income as Low, Average, or High based on quartiles within their state and year. This would allow analysis of chamber or district population across the entire dataset but reduce data granularity and statistical power.

H1: Higher-income districts have more effective legislators

Regression Analysis	
	<i>Dependent variable:</i>
	State Legislative Effectiveness Score
Female	-0.092*** (0.018)
Worker	-0.049 (0.031)
Minority Race	-0.075*** (0.029)
Seniority	0.044*** (0.002)
District Per Capita Mean Income	0.00001*** (0.00000)
Observations	14,238
R ²	0.043
Adjusted R ²	0.024
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 Note: Standard errors in parentheses.	

Figure 2: Regression analysis of SLES on District Per Capita Mean Income and covariates using fixed effects

I regress the State Legislative Effectiveness Score on district per capita mean income and additional explanatory variables of female, worker, minority race, and seniority and include fixed effects accounting for state, chamber, and year.

Consistent with Lollis' findings, being a worker does not appear to affect a legislator's SLES. Increasing district per capita mean income has a statistically significant effect on increasing that district legislator's SLES albeit by a small amount, especially when compared to the other legislator effects.

H2: Higher-income districts have fewer worker legislators

Regression Analysis	
	<i>Dependent variable:</i>
	Worker
SLES	-0.004 (0.002)
Female	-0.039*** (0.005)
Minority Race	-0.011 (0.008)
Seniority	-0.0005 (0.001)
District Per Capita Mean Income	-0.00000*** (0.00000)
Observations	14,238
R ²	0.033
Adjusted R ²	0.014
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01 Note: Standard errors in parentheses.

Figure 3: Regression analysis of Worker status on District Per Capita Mean Income and covariates using fixed effects

I regress worker status on district per capita mean income and additional covariates of SLES, female, minority race, and seniority and include fixed effects accounting for state, chamber, and year. Increasing district per capita mean income has a statistically significant effect on decreasing the likelihood that a legislator is a worker, albeit by a small amount, especially when compared to the effect of being female.

H3: Higher-income districts are more likely to remove legislators

Regression Analysis	
	<i>Dependent variable:</i>
	Re-Elected
SLES	0.023*** (0.004)
Female	0.0004 (0.009)
Worker	-0.006 (0.016)
Minority Race	0.006 (0.015)
District Per Capita Mean Income	-0.00000** (0.00000)
Observations	10,321
R ²	0.363
Adjusted R ²	0.350
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01 Note: Standard errors in parentheses.

Figure 4: Regression analysis of reelected status on SLES, District Per Capita Mean Income, and covariates using fixed effects

I regress whether reelected on SLES, worker, and district per capita mean income and additional covariates of female and minority race and include fixed effects accounting for state, chamber, and year. Being working-class does not have a statistically significant effect on a legislator being reelected, but this does not account for self-selection amongst working-class candidates choosing to run for election as Lollis (2024) points out. Increased SLES is associated with increased reelection while increased district per capita mean income is associated with decreased reelection, albeit by a very small amount. This finding could point to higher-income districts being more active in assessing their legislators' effectiveness and responding by removing them from office.

Conclusion

Higher-income districts behave differently from lower-income districts when it comes to electing and reelecting legislators. Higher-income districts have more effective legislators, fewer working-class legislators, and are more likely to remove legislators. These differences could result in different legislative outcomes and affect citizen quality of life especially over the long term.

Future research

North Dakota's House of Representatives District 31's experience with Representative Rohr is broadly illustrative of this analysis' findings but asks additional questions that would benefit from future research. Representative Rohr is a female, non-worker legislator in a low-income district who is ineffective according to the SLES indicator. If this district were higher-income, then perhaps she would be more effective or she would not be reelected as higher-income districts tend to have more effective legislators and remove legislators slightly more often. Perhaps higher-income districts have a more effective legislator candidate pool or perhaps voters and donors in higher-income districts lobby their legislators more actively and contribute to increasing their effectiveness. While low-income districts tend to have more worker legislators, that Representative Rohr was elected and continues to be reelected suggests that perhaps other confounding effects are stronger. Low-income districts tend to reelect more often or take longer on average to remove legislators, which could be explained by voters being less informed or politically engaged or incumbents having a stronger advantage in lower-income districts.

Additional analysis could investigate potential covariates such as district age distribution or race and its relationship to district income. District income distribution, instead of district per capita mean income, could be used to account for inequality as well as absolute level, such as is available in the U.S. Census Bureau ACS Table S2901 for years 2017-2023 at the federal Congressional level. Analyzing the income distribution could provide insight into whether the level of inequality affects the power of voters and political inequality and whether the absolute level of income affects the political priorities, as Griffin and Flavin indicate. Per capita mean income cannot differentiate between a highly unequal distribution and an even distribution, but these two distributions could result in different district legislators and political outcomes.

Implications

This analysis suggests potential implications for the SLES measure and redistricting practices across the U.S. The SLES is being used nationally for training and proving influential amongst political donors and legislators. The Center of Effective Lawmaking recently partnered with the Arthur M. Blank

Family Foundation to publicize the SLES indicator and lobby state legislators to improve their SLES and lawmaking effectiveness through pilot programs starting in Georgia and Montana. However, this focus on the SLES could bias against low-income districts and inadvertently penalize them. Griffin and Flavin (2011) find that different types of districts and their voters prefer their legislators to focus on different types of work; white or higher-income voters prioritize policy representation, but minority or lower-income voters prioritize receiving funding for their districts. They find that legislators are responsive to these priorities and focus their energies according to their voters' preferences. If this continues to be the case today, then it's possible that since the SLES measure doesn't give direct credit for appropriations success underestimates the effectiveness of legislators from districts with significant minority or lower-income voter populations. As the SLES measure starts being used to guide legislators and foundations looking to improve their states' governance, this underestimation could lead donors who focus on increasing SLES indicators to incentivize legislators to shift focus from funding concerns to legislative advancement, potentially putting them at odds with their voters. This analysis suggests that the SLES could benefit from including appropriations success as a measure of legislative effectiveness, updating Jeydel and Taylor's indicator.

There could also be practical implications for redistricting, such as suggesting whether additional demographic factors should be considered in redistricting or at least provided in the U.S. Census' redistricting data file. The current Census standard redistricting file provides demographic data on race, adult population, and special populations, such as military and prison populations. States vary in how they use this data and what additional data, such as holistic communities of interest, they consider in their redistricting. The only federal requirements are that population distribution and racial distribution must be fair in accordance with the Equal Protection Clause of the U.S. Constitution (The American Redistricting Project). This analysis shows that district income level has a significant effect on whether districts are able to successfully achieve their political goals through electing and reelecting effective legislators to represent them. Higher-income districts tend to have more effective legislators, fewer worker legislators, and remove legislators more often, potentially suggesting that they are more politically engaged and responsive to their legislators' effectiveness. This systematic difference in political success has long term implications for districts and suggests the need to investigate a systematic adjustment. Future research could provide for updated methodologies for the SLES measure and redistricting to allow for equal opportunity for legislative effectiveness.

References

- Bratton, K. A. (2006). The Behavior and Success of Latino Legislators: Evidence from the States. *Social Science Quarterly*, 87(5), 1136–1157.
- Bucchianeri P., Volden C., Wiseman A. E. (2024). Legislative Effectiveness in the American States. *American Political Science Review*. Published online 2024:1-19.
- Carnes, N. (2012). Does the Numerical Underrepresentation of the Working Class in Congress Matter? *Legislative Studies Quarterly*, 37(1), 5–34.
- Carnes, Nicholas. (2013). *White-Collar Government: The Hidden Role of Class in Economic Policy Making*. Chicago, IL: University of Chicago Press.
- Canon, D. T. (2022). Race and Redistricting. *Annual Review of Political Science*, (25), 509-528.
- Gerrity, J. C., Osborn, T., & Mendez, J. M. (2007). Women and Representation: A Different View of the District? *Politics & Gender*, 3(2), 179–200.
- Griffin, J. D., & Flavin, P. (2011). How Citizens and Their Legislators Prioritize Spheres of Representation. *Political Research Quarterly*, 64(3), 520-533.
- Hunt, C. R. & Miler, K. C. (2024). How Modern Lawmakers Advertise Their Legislative Effectiveness to Constituents. *The Journal of Politics*. Forthcoming.
- Jeydel, A., & Taylor, A. J. (2003). Are Women Legislators Less Effective? Evidence from the U.S. House in the 103rd-105th Congress. *Political Research Quarterly*, 56(1), 19–27.
- Lollis, J. M. (2024). Are Workers Effective Lawmakers? *State Politics & Policy Quarterly*, 1–17.
- Mansbridge, J. (1999). Should Blacks Represent Blacks and Women Represent Women? A Contingent “Yes.” *The Journal of Politics*, 61(3), 628-657.
- Miquel, G. P. I., & Snyder, J. M. (2006). Legislative Effectiveness and Legislative Careers. *Legislative Studies Quarterly*, 31(3), 347–381.
- Nickelson, J., & Jansa, J. M. (2023). Descriptive Representation and Innovation in American Legislatures. *Political Research Quarterly*, 76(4), 2018-2035.

U.S. Census Bureau. (2010, 2012, 2014, 2016). MEAN INCOME IN THE PAST 12 MONTHS (IN 2010 INFLATION-ADJUSTED DOLLARS) . *American Community Survey, ACS 5-Year Estimates Subject Tables, Table S1902*.

Volden, C., Wiseman, A.E. and Wittmer, D.E. (2013), When Are Women More Effective Lawmakers Than Men?. *American Journal of Political Science*, 57: 326-341.

Weissert, C. S. (1991). Issue Salience and State Legislative Effectiveness. *Legislative Studies Quarterly*, 16(4), 509–520.